

Empirical Exercises for Economics of Global Shifts Course

1. Global Trade and Economic Growth

Objective: Analyze the relationship between global trade and economic growth in different countries.

Instructions:

- Collect data on GDP and trade volume (exports + imports) for a selection of countries over the past 30 years (you can use databases like the World Bank, IMF, or UNCTAD).
- Plot the trade-to-GDP ratio and growth rates for each country over time.
- Perform a correlation analysis between trade volume and GDP growth for each country.
- Discuss the results: Do higher levels of trade seem to correlate with higher economic growth? Are there differences between developed and developing countries?

Empirical Tasks:

1. Compute the trade-to-GDP ratio for each country.
2. Run a regression analysis to determine the relationship between trade and GDP growth.
3. Evaluate whether there are time-lags between trade expansion and GDP growth.

2. Technological Advancements and Productivity Growth

Objective: Investigate the role of technological advancements (e.g., ICT, AI) in productivity growth.

Instructions:

- Use a dataset such as the OECD’s productivity database or World Bank data on labor productivity and technological innovation indicators.
- Choose a few countries (e.g., the U.S., China, India, and Germany) and gather data on productivity (output per hour worked) and technological investments (R&D expenditure, patents granted).
- Analyze whether higher levels of technological innovation are associated with higher productivity growth.

Empirical Tasks:

1. Calculate labor productivity growth rates for each country over the last 10 years.
2. Run a regression analysis to see if there is a statistically significant relationship between technological innovation (measured by R&D spending or patents) and productivity growth.
3. Compare the results across countries with different levels of technological development.

3. Automation and Employment

Objective: Examine the impact of automation on labor markets, especially in manufacturing sectors.

Instructions:

- Collect data on automation adoption rates in key industries (e.g., manufacturing, retail) and employment data (total jobs and sector-specific jobs) from the Bureau of Labor Statistics (U.S.) or national labor data sources.
- Focus on countries with varying levels of automation, such as the U.S., South Korea, and Germany.
- Analyze the impact of automation on employment trends: Are jobs being displaced, or is there a shift to higher-skill employment?

Empirical Tasks:

1. Calculate the change in employment levels in automation-heavy industries over the last 20 years.
2. Use a difference-in-differences (DiD) approach to compare regions with varying automation adoption rates.
3. Interpret the impact of automation on job displacement and wage levels in manufacturing versus service sectors.

4. Foreign Direct Investment (FDI) and Economic Growth

Objective: Investigate the role of foreign direct investment (FDI) in stimulating economic growth in developing countries.

Instructions:

- Use the World Bank or UNCTAD database to gather data on FDI inflows and GDP growth for a set of developing countries (e.g., Brazil, India, South Africa).
- Examine how changes in FDI inflows are correlated with GDP growth and other economic indicators such as employment and infrastructure development.

Empirical Tasks:

1. Create a time series plot of FDI inflows and GDP growth for each country.
2. Perform a regression analysis to see if there is a statistically significant impact of FDI on GDP growth.
3. Compare the growth rates of countries with high FDI inflows to those with low FDI inflows and discuss the potential reasons for observed differences.

5. Digital Economy and Inequality

Objective: Assess the effect of the digital economy on income inequality across countries.

Instructions:

- Collect data on internet penetration rates, digital services usage (e.g., e-commerce, digital banking), and income inequality (Gini coefficient) for a set of countries over the last 20 years.
- Investigate whether there is a relationship between the expansion of digital services and changes in income inequality.

Empirical Tasks:

1. Compute the Gini coefficient for a selection of countries over time and compare it with the level of digital penetration.
2. Run a regression model to test whether increased digital services usage is associated with lower or higher income inequality.
3. Discuss the potential channels through which the digital economy might affect inequality (e.g., access to education, job creation, wealth accumulation).

6. Global Supply Chains and Manufacturing Relocation

Objective: Explore the effects of globalization and global supply chains on manufacturing locations and employment.

Instructions:

- Use data on global manufacturing output, trade flows, and employment (e.g., from the WTO or World Bank).
- Examine the movement of manufacturing jobs from high-wage countries to low-wage countries over the past few decades (e.g., U.S. to China, Mexico to Central America).
- Analyze the effects on wages, employment rates, and regional economic development.

Empirical Tasks:

1. Plot trends in manufacturing output and employment in major countries over time.
2. Use a panel data regression model to assess how changes in global supply chain integration (e.g., outsourcing, offshoring) correlate with domestic manufacturing employment and wages.
3. Evaluate the impact on income inequality within countries affected by offshoring.

7. Environmental Impact of Technological Change

Objective: Analyze the environmental consequences of technological shifts in key industries.

Instructions:

- Collect data on energy consumption, carbon emissions, and technological advancements in energy efficiency or green technologies.
- Focus on industries such as energy production, manufacturing, and transportation.
- Examine the environmental effects of technological advancements in these sectors and their relationship to economic growth.

Empirical Tasks:

1. Analyze the relationship between technological innovation in renewable energy (e.g., wind, solar) and reductions in carbon emissions.

2. Calculate the carbon intensity of GDP (CO2 emissions per unit of GDP) for countries with varying levels of technological adoption.
3. Perform a regression analysis to determine the effect of green technologies on reducing environmental degradation without sacrificing economic growth.

8. Demographic Shifts and Economic Outcomes

Objective: Investigate how demographic changes (e.g., aging populations) influence economic outcomes such as growth, savings rates, and labor force participation.

Instructions:

- Use demographic data from the UN Population Division and economic data (GDP growth, savings rates, labor force participation) from the World Bank.
- Focus on countries with rapidly aging populations (e.g., Japan, Germany) and compare them to countries with younger populations (e.g., India, Brazil).

Empirical Tasks:

1. Analyze the relationship between the aging population and GDP growth rates across countries.
2. Perform a regression analysis to assess the impact of demographic aging on national savings rates and labor force participation.
3. Discuss potential policy responses to mitigate the economic effects of demographic shifts, such as pension reform or changes in immigration policy.